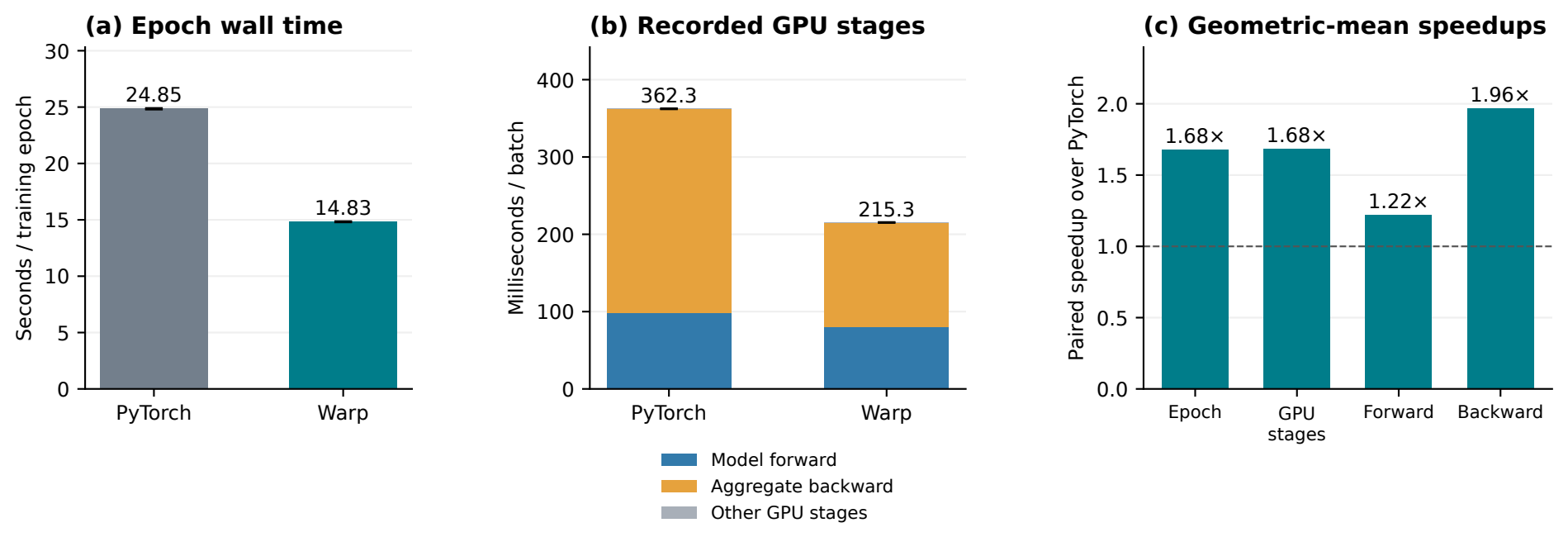
(a) Epoch wall time
Seconds / training epoch
30
25
20
15
10
5
0
24.85
14.83
PyTorch
Warp
(b) Recorded GPU stages
Milliseconds / batch
400
300
200
100
0
362.3
215.3
PyTorch
Warp
Model forward
Aggregate backward
Other GPU stages
(c) Geometric-mean speedups
Paired speedup over PyTorch
2.0
1.5
1.0
0.5
0.0
1.68×
1.68×
1.22×
1.96×
Epoch
GPU stages
Forward
Backward